YCCE CT App

System Overview — Admin & Faculty Reference

What is it?

A full-stack college management system built for the Computer Technology department at YCCE. It has two parts — a mobile app for students and faculty, and a web-based admin dashboard for administrators and professors.

TECHNOLOGIES USED

Backend

Node.js — Runs the server
Express.js — Handles routes and API endpoints
PostgreSQL — Stores all app data
node-postgres (pg) — Connects Node.js to PostgreSQL
JWT (jsonwebtoken) — Issues and verifies login tokens
bcryptjs — Hashes and verifies passwords
Multer — Parses file uploads from requests
Cloudinary — Stores uploaded files (notes, assignments) on the cloud
Nodemailer — Sends OTP emails for verification and password reset
node-cron — Runs scheduled tasks every minute/hour automatically
helmet — Adds security headers to every response
cors — Allows the web dashboard and app to talk to the backend
morgan — Logs every incoming request in the terminal
dotenv — Loads secret keys and config from the .env file

Mobile App

React Native — Builds the UI for both iOS and Android from one codebase
Expo SDK 54 — Simplifies building, running, and deploying the app
expo-notifications — Requests permission and receives push notifications
expo-document-picker — Lets users pick files from their device to upload
AsyncStorage — Saves login token locally so users stay logged in
axios — Makes HTTP requests to the backend API
React Navigation — Moves between screens (login, home, subject detail, etc.)
Ionicons — Renders icons throughout the app
React Context API — Shares auth and theme state across all screens globally

Admin Web Dashboard

Next.js 16 — Renders the dashboard pages with React
TypeScript — Catches type errors at compile time
Tailwind CSS — Styles every component using utility classes
shadcn/ui — Provides ready-made components like dialogs, tables, and cards
lucide-react — Renders icons across all dashboard pages
next-themes — Toggles dark and light mode across the entire site
axios — Sends API requests from the dashboard to the backend

Services & Infrastructure

Cloudinary — Hosts uploaded files and serves them via CDN link
Expo Push Service — Delivers notifications to student devices
PostgreSQL — Single source of truth for all data
Node HTTPS — Calls the Expo Push API directly from the backend cron job

SYSTEM OVERVIEW

Registration & Security

- Students can only register with college email IDs ending in @ycce.in
- The email prefix must match their roll number format
- New student accounts require admin approval before they can log in
- Passwords are stored as bcrypt hashes — never in plain text
- Every API request is protected with a JWT Bearer token

Mobile App — Students & Faculty

Subjects — Each student sees subjects for their year with 4 tabs: Announcements, Notes, Assignments, Attendance

Announcements — Faculty post subject-specific announcements; students receive them under the subject

Notes — Faculty upload PDFs with title and unit tag; students view and download directly

Assignments — Faculty create assignments with due date and file; students submit notes or files

Attendance — Students see per-subject attendance with a mini calendar (green = present, red = absent)

Push Notifications — Students get alerts 15 min before class and 24 hours before assignment deadline

Admin Web Dashboard — Professors & Admins

Overview — Live stats cards (Total Students, Approved, Pending, Announcements) — each clickable to expand full list

Users — Approve or reject student registrations; view all students in a table

Announcements — Post general or year-specific announcements with type badge

Attendance — Mark attendance per subject per date with calendar view, search bar, and bulk P/A buttons

Results — Add or update SGPA per student per semester with auto-filled semester and academic year

Timetable — Add lectures with custom subject name, day, time, and room — filtered by year

Assignments — Create assignments with file attachment; view all student submissions per assignment

Faculty — View all faculty accounts with name, email, and role

Dark / Light Mode — Toggle in the sidebar footer

NOTIFICATION SYSTEM

Trigger	Recipients	Message
15 min before lecture	Students of that year	"Subject starts in 15 minutes"
Lecture starts now	Students of that year	"Your lecture has begun"
24 hrs before deadline	Students who have not submitted	"Assignment due tomorrow"

DATABASE TABLES (POSTGRESQL)

users · subjects · assignments · assignment_submissions · attendance · timetable · announcements · notes · sgpa_results · notification_logs

All relationships use foreign keys. Unique constraints prevent duplicate attendance records and submissions. Upsert (ON CONFLICT DO UPDATE) is used for attendance and SGPA so re-submitting updates the existing record.

KEY ARCHITECTURAL PATTERNS

REST API — All communication between frontend and backend uses HTTP endpoints

JWT Auth — Every protected route validates the Bearer token before responding

Role-based Access — Students, professors, and admins each get different data and permissions

Multipart/form-data — Sends files alongside text in a single HTTP request

Cron Jobs — Automatically notify students without any manual trigger

Upsert Pattern — Updates existing records instead of throwing errors for duplicates